: Laboratory of Intelligent Systems

# Automated Aerodynamic Characterization of a Morphing VTOL

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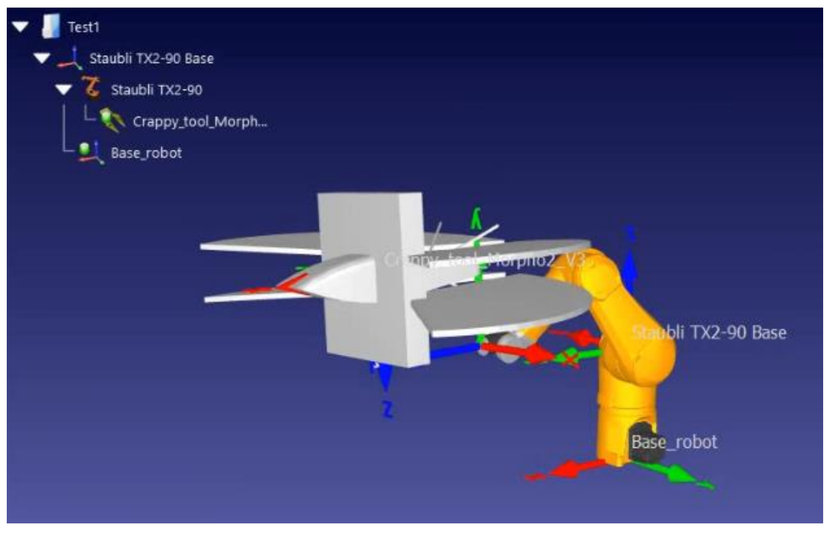
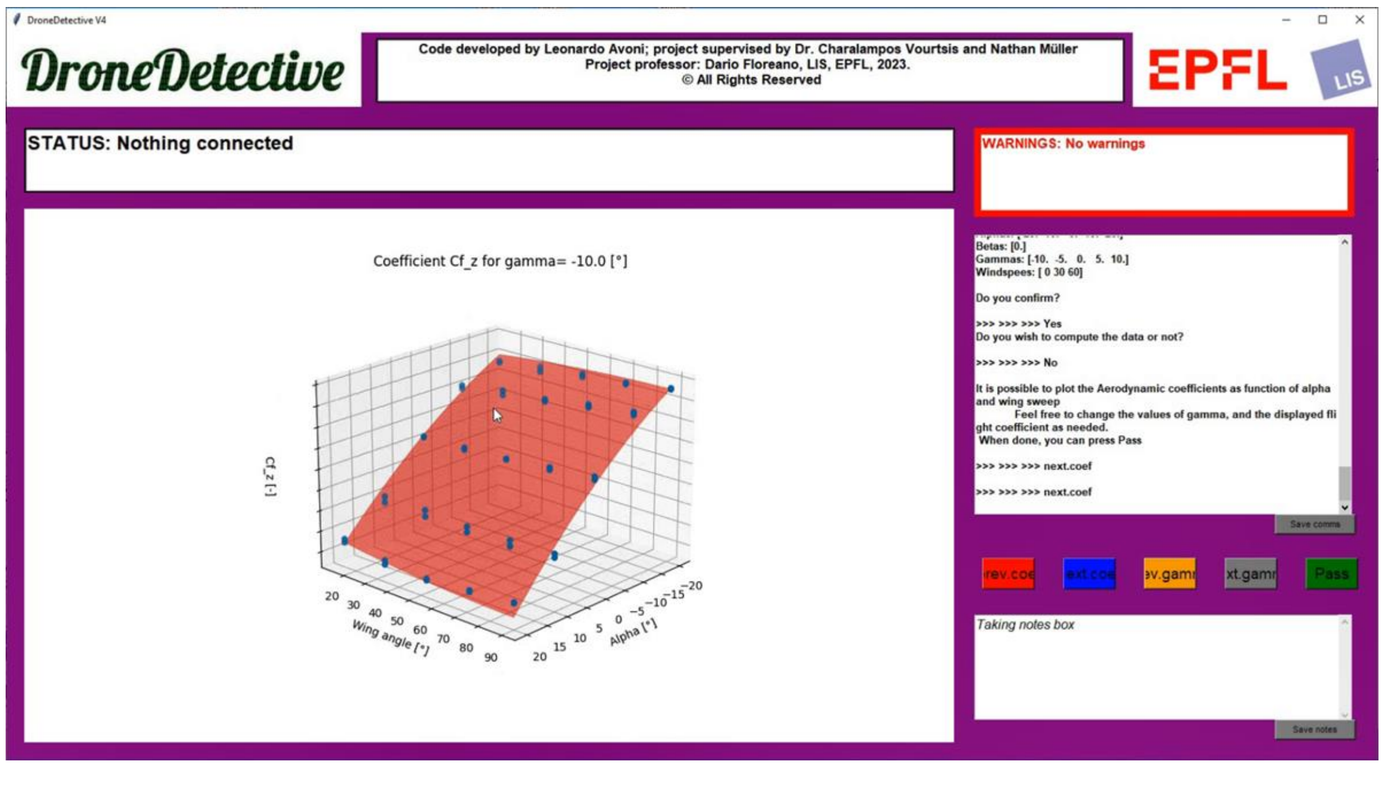
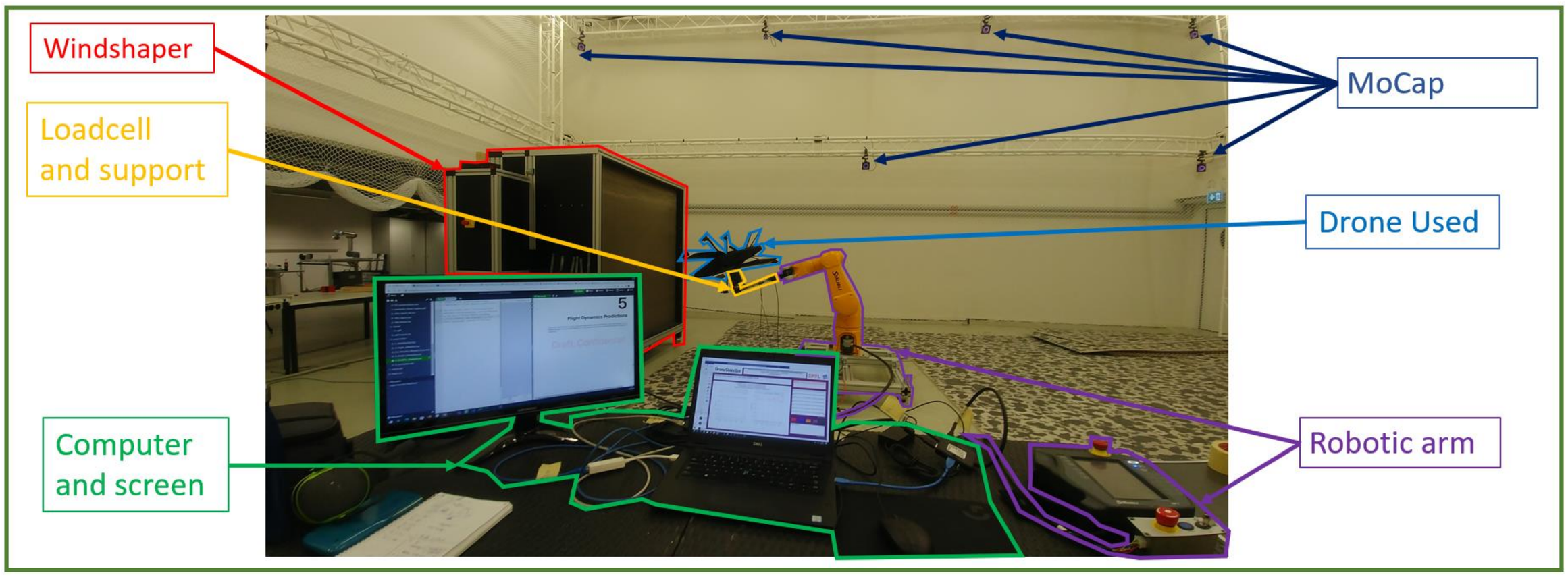
The project was aimed at building a tool, called DroneDetective, used to characterize automatically the aerodynamics of (morphing) drones. DroneDetective can extract data about:

- The aerodynamics of the drone
- The center of gravity of the drone
- Vibrations of the drone




DroneDetective can be used both for recording automatically the needed wind-tunnel data, and for the automatic data analysis. The Hardware Setup was composed of a robotic arm, a Windshaper, a MoCap system, a loadcell and the morphing drone to be tested




DroneDetective's operations allowed to cut the experimentation times by a factor 10, while also increasing repeatability and accuracy. Operator errors can thus be avoided

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